



**NORTH SYDNEY GIRLS HIGH SCHOOL
YEAR 12 – TERM 2 ASSESSMENT
2006**

**MATHEMATICS
EXTENSION COURSE 2**

TIME ALLOWED: 60 minutes
Plus 2 minutes reading time

INSTRUCTIONS:

- Start each question on a new page
- Hand each question in separately, including a sheet for non-attempts
- Show all necessary working

This task is worth 32% of the HSC Assessment Mark

Question 1 (21 Marks)

a) Find :

i. $\int (\cos x + \sin x) e^{(\cos x - \sin x)} dx$ 2

ii. $\int \tan^3 x \, dx$ 3

iii. $\int \cos^{-1} x \, dx$ 3

b)

i. Find real constants a , b and c such that

$$\frac{1}{(x^2 + 1)(x + 1)} \equiv \frac{ax + b}{x^2 + 1} + \frac{c}{x + 1} \quad 3$$

ii. Hence evaluate $\int_0^1 \frac{dx}{(x^2 + 1)(x + 1)}$ 4

c) If $I_n = \int \sin^n x \, dx$ for $n \geq 0$,

i. Show that $I_n = -\frac{1}{n} \cos x \sin^{n-1} x + \frac{n-1}{n} I_{n-2}$ for $n \geq 2$. 4

ii. Hence evaluate $\int_{\pi}^{3\pi} \sin^4 x \, dx$ 2

Question 2 (17 Marks)

a)

- i. Show that the equation of the normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point (x_1, y_1) is $\frac{xa^2}{x_1} + \frac{yb^2}{y_1} = a^2 + b^2$. **3**

- ii. The normal to the hyperbola $\frac{x^2}{2} - y^2 = 1$ at $P\left(\sqrt{3}, \frac{1}{\sqrt{2}}\right)$ cuts the y -axis at A and the x -axis at B . Show that $PA:PB = 2:1$. **4**

- b) The chord of contact from an external point R to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ meets the ellipse at P and Q .

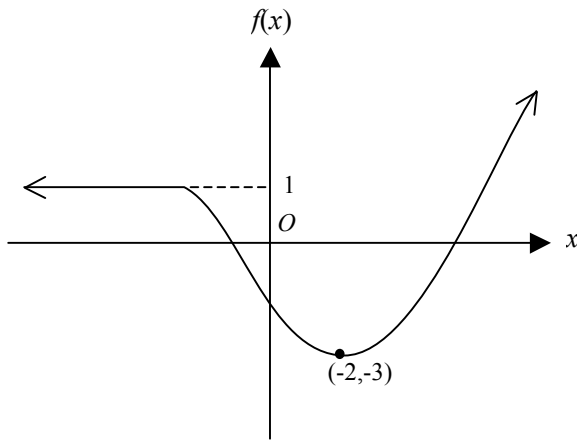
If R lies on the directrix $x = \frac{a}{e}$, show that PQ is a focal chord. **3**

- c) The ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ has centre O . P is the point on this ellipse with parameter θ .

- i. Find the gradient of the tangent at P . **1**
- ii. A line drawn through O , parallel to the tangent to the ellipse at P , meets the ellipse at Q and R . Find the coordinates of Q and R . **3**
- iii. Prove that the area of triangle PQR is independent of the position of P . **3**

Question 3 (22 Marks)

a) Here is a graph of $y = f(x)$:



i. For what value(s) of k does the equation $f(x) = k$ have no solutions? **1**

ii. For what value(s) of k does the equation $f(x) = k$ have an infinite number of solutions? **1**

b) Consider the relation $x^2 + y^2 + bxy = 1$

Suppose that $P(x_1, y_1)$ is a point on the graph of this relation where the tangent to the graph is vertical.

i. Show that $y_1 = \frac{-bx_1}{2}$ **3**

ii. Hence find the values of b for which the graph of $x^2 + y^2 + bxy = 1$ contains points at which the tangent to the graph is vertical. **4**

Question 3 continued

- c) The equation of the tangent at $P\left(ct, \frac{c}{t}\right)$ on the rectangular hyperbola $xy = c^2$ is $x + t^2y = 2ct$.

i. Find the equation of the perpendicular to this tangent passing through the origin.

1

ii. Let T be the point where the line in part (i) intersects the tangent.

Find the equation of the locus of T .

5

d)

i. Show that $\int_a^b f(x) dx = \int_a^b f(b-x+a) dx$

3

ii. Prove that $\int_L^{L+\frac{\pi}{2}} \frac{A \sin(x-L) + A + B \cos(x-L) + B}{\sin(x-L) + \cos(x-L) + 2} dx = \frac{\pi}{4} (A+B)$

4

where L , A and B are constants

End of paper

Solutions

$$(1) \quad (a) \quad (i) \quad \int (\cos x + \sin x) e^{(\cos x - \sin x)} dx = - \int (-\sin x - \cos x) e^{(\cos x - \sin x)} dx \\ = -e^{(\cos x - \sin x)} + c$$

$$(ii) \quad \int \tan^3 x dx = \int \tan x \cdot \tan^2 x dx \\ = \int \tan x (\sec^2 x - 1) dx \\ = \int \tan x \sec^2 x dx - \int \tan x dx \\ = \frac{\tan^2 x}{2} - \int \frac{\sin x dx}{\cos x} \\ = \frac{\tan^2 x}{2} + \int \frac{-\sin x dx}{\cos x} \\ = \frac{\tan^2 x}{2} + \ln |\cos x| + c$$

$$(iii) \quad \int \cos^{-1} x dx = \int 1 \times \cos^{-1} x dx \\ = \int \frac{d}{dx}(x) \times \cos^{-1} x dx \\ = x \cos^{-1} x - \int x \times \left(-\frac{1}{\sqrt{1-x^2}} \right) dx \\ = x \cos^{-1} x - \frac{1}{2} \int \left(\frac{-2x dx}{\sqrt{1-x^2}} \right) \\ = x \cos^{-1} x - \frac{1}{2} \int \left(\frac{1}{\sqrt{u}} \right) du \\ = x \cos^{-1} x - \frac{1}{2} [2\sqrt{u}] + c \\ = x \cos^{-1} x - \sqrt{1-x^2} + c$$

$$(b) \quad (i) \quad \frac{1}{(x^2+1)(x+1)} \equiv \frac{ax+b}{x^2+1} + \frac{c}{x+1} \\ \therefore 1 = (x+1)(ax+b) + c(x^2+1) \\ x = -1: \quad 1 = 2c \Rightarrow c = \frac{1}{2} \\ x = 0: \quad 1 = b + c \Rightarrow b = \frac{1}{2} \\ a + c = 0 \quad \left[\text{equating coefficient of } x^2 \right] \\ \therefore a = -\frac{1}{2} \\ \therefore a = -\frac{1}{2}, b = \frac{1}{2}, c = \frac{1}{2}$$

$$\begin{aligned}
\text{(ii)} \quad \int_0^1 \frac{dx}{(x^2+1)(x+1)} &= \frac{1}{2} \int \left(\frac{-x+1}{x^2+1} + \frac{1}{x+1} \right) dx \\
&= \frac{1}{2} \int \left(\frac{-x}{x^2+1} \right) dx + \frac{1}{2} \int \left(\frac{1}{x^2+1} \right) dx + \frac{1}{2} \int \left(\frac{1}{x+1} \right) dx \\
&= -\frac{1}{4} \int \left(\frac{2x}{x^2+1} \right) dx + \frac{1}{2} \tan^{-1} x + \frac{1}{2} \ln|x+1| + c \\
&= -\frac{1}{4} \ln(x^2+1) + \frac{1}{2} \tan^{-1} x + \frac{1}{2} \ln|x+1| + c
\end{aligned}$$

$$\begin{aligned}
\text{(c)} \quad \text{(i)} \quad I_n &= \int \sin^n x \, dx \\
&= \int \sin^{n-1} x \times \sin x \, dx \\
&= \int \sin^{n-1} x \times \frac{d}{dx}(-\cos x) \, dx \\
&= -\cos x \sin^{n-1} x - \int -\cos x \times \frac{d}{dx}(\sin^{n-1} x) \, dx \\
&= -\cos x \sin^{n-1} x + (n-1) \int \cos^2 x \sin^{n-2} x \, dx \\
&= -\cos x \sin^{n-1} x + (n-1) \int (1 - \sin^2 x) \sin^{n-2} x \, dx \\
&= -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x \, dx - (n-1) \int \sin^n x \, dx \\
&= -\cos x \sin^{n-1} x + (n-1) I_{n-2} - (n-1) I_n \\
\therefore I_n &= -\cos x \sin^{n-1} x + (n-1) I_{n-2} - (n-1) I_n \\
\therefore I_n + (n-1) I_n &= n I_n = -\cos x \sin^{n-1} x + (n-1) I_{n-2} \\
\therefore I_n &= -\frac{1}{n} \cos x \sin^{n-1} x + \left(\frac{n-1}{n} \right) I_{n-2}
\end{aligned}$$

$$\begin{aligned}
\text{(ii)} \quad \int_{\pi}^{3\pi} \sin^4 x \, dx &= I_4 \\
&= \left[-\frac{1}{4} \cos x \sin^3 x \right]_{\pi}^{3\pi} + \frac{3}{4} I_2 \\
&= \frac{3}{4} I_2 \\
&= \frac{3}{4} \left[\left[-\frac{1}{2} \cos x \sin x \right]_{\pi}^{3\pi} + \frac{1}{2} I_0 \right] \\
&= \frac{3}{8} I_0 \\
&= \frac{3}{8} \int_{\pi}^{3\pi} (\sin x)^0 \, dx \\
&= \frac{3}{8} [3\pi - \pi] \\
&= \frac{3\pi}{4}
\end{aligned}$$

$$(2) \quad (a) \quad (i) \quad \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\frac{2x}{a^2} - \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\therefore \frac{dy}{dx} = \frac{b^2 x}{a^2 y}$$

$$\text{At } (x_1, y_1), \frac{dy}{dx} = \frac{b^2 x_1}{a^2 y_1}$$

$$\therefore \text{Normal has gradient } -\frac{a^2 y_1}{b^2 x_1}$$

$$y - y_1 = -\frac{a^2 y_1}{b^2 x_1} (x - x_1)$$

$$\therefore b^2 x_1 y - b^2 x_1 y_1 = -a^2 y_1 x + a^2 y_1 x_1$$

$$\therefore a^2 y_1 x + b^2 x_1 y = a^2 y_1 x_1 + b^2 x_1 y_1 = x_1 y_1 (a^2 + b^2)$$

$$\therefore \frac{xa^2}{x_1} + \frac{yb^2}{y_1} = a^2 + b^2 \quad [\div x_1 y_1]$$

$$(ii) \quad \frac{x^2}{2} - y^2 = 1 \Rightarrow a^2 = 2, b^2 = 1$$

$$(x_1, y_1) = \left(\sqrt{3}, \frac{1}{\sqrt{2}} \right)$$

$$\text{So the normal has equation } \frac{2x}{\sqrt{3}} + \sqrt{2}y = 3$$

$$A: \left(0, \frac{3}{\sqrt{2}} \right)$$

$$B: \left(\frac{3\sqrt{3}}{2}, 0 \right)$$

$$PA^2 = (\sqrt{3})^2 + \left(\frac{1}{\sqrt{2}} - \frac{3}{\sqrt{2}} \right)^2 = 3 + \left(\frac{2}{\sqrt{2}} \right)^2 = 5$$

$$PB^2 = \left(\sqrt{3} - \frac{3\sqrt{3}}{2} \right)^2 + \left(\frac{1}{\sqrt{2}} \right)^2 = \left(\frac{2\sqrt{3} - 3\sqrt{3}}{2} \right)^2 + \frac{1}{2} = \frac{3+2}{4} = \frac{5}{4}$$

$$\therefore PA^2 : PB^2 = 5 : \frac{5}{4} = 4 : 1$$

$$\therefore PA : PB = 2 : 1 (= a^2 : b^2)$$

$$(b) \quad \text{If } R(x_1, y_1) \text{ then the chord of contact has equation } \frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$$

$$x_1 = \frac{a}{e} : \frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1 \Rightarrow \frac{x}{ae} + \frac{yy_1}{b^2} = 1$$

The point $(ae, 0)$ clearly lies on this chord (check $y = 0$), so it is a focal chord!

(c) (i) $P(a \cos \theta, b \sin \theta)$

$$\frac{2x}{a^2} + \frac{2y}{b^2} y' = 0$$

$$\therefore y' = -\frac{b^2 x}{a^2 y}$$

$$\therefore y'_P = -\frac{b^2 a \cos \theta}{a^2 b \sin \theta} = -\frac{b \cos \theta}{a \sin \theta}$$

(ii) $y = -\frac{b \cos \theta}{a \sin \theta} x$ is the line parallel to the tangent passing through O

$$y = -\frac{b \cos \theta}{a \sin \theta} x \Rightarrow \frac{y}{b} = -\frac{\cos \theta}{a \sin \theta} x = -\frac{1}{a} \cot \theta$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \Rightarrow \frac{x^2}{a^2} + \left(-\frac{x}{a} \cot \theta\right)^2 = 1$$

$$\therefore \frac{x^2}{a^2} (1 + \cot^2 \theta) = 1$$

$$\therefore \frac{x^2}{a^2} (\operatorname{cosec}^2 \theta) = 1 \Rightarrow x = \pm a \sin \theta$$

$$\therefore y = \mp b \cos \theta$$

$$Q(a \sin \theta, -b \cos \theta), R(-a \sin \theta, b \cos \theta)$$

(iii) $y = -\frac{b \cos \theta}{a \sin \theta} x \Rightarrow xb \cot \theta + ay = 0$

Let d be the distance of P from this line

$$\begin{aligned} d &= \frac{|a \cos \theta \times b \cot \theta + a(b \sin \theta)|}{\sqrt{b^2 \cot^2 \theta + a^2}} = \frac{\frac{ab \cos^2 \theta + ab \sin^2 \theta}{|\sin \theta|}}{\sqrt{b^2 \cot^2 \theta + a^2}} \\ &= \frac{ab}{|\sin \theta| \sqrt{b^2 \cot^2 \theta + a^2}} = \frac{ab}{\sqrt{b^2 \sin^2 \theta \cot^2 \theta + a^2 \sin^2 \theta}} \\ &= \frac{ab}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}} \end{aligned}$$

$$QR^2 = (2a \sin \theta)^2 + (-2b \cos \theta)^2 = 4(a^2 \sin^2 \theta + b^2 \cos^2 \theta)$$

$$\therefore QR = 2\sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}$$

$$\text{Area } \Delta PQR = \frac{d}{2} \times QR = ab$$

This area is independent of the position of P .

- (3) (a) (i) $k < -3$
(ii) $k = 1$

(b) (i) $x^2 + y^2 + bxy = 1$
 $\therefore 2x + 2yy' + b(xy' + y) = 0$
 $\therefore 2yy' + bxy' = -(2x + by)$
 $\therefore (2y + bx)y' = -(2x + by)$
 $\therefore y' = -\frac{2x + by}{2y + bx}$

Vertical tangents are when y' is undefined ie $2y + bx = 0$

$$\therefore 2y_1 + bx_1 = 0$$

$$\therefore y_1 = -\frac{bx_1}{2}$$

(ii) $x_1^2 + y_1^2 + bx_1y_1 = 1$

$$y_1 = -\frac{bx_1}{2}$$

$$\therefore x_1^2 + \left(-\frac{bx_1}{2}\right)^2 + bx_1\left(-\frac{bx_1}{2}\right) = 1$$

$$\therefore 4x_1^2 + b^2x_1^2 - 2b^2x_1^2 = 4$$

$$\therefore 4x_1^2 - b^2x_1^2 = 4$$

$$\therefore (4 - b^2)x_1^2 = 4$$

This will have a solution as long as $4 - b^2 > 0 \Rightarrow -2 < b < 2$

- (c) (i) $Ax + By + C = 0$ has perpendicular $Bx - Ay + D = 0$
 $\therefore x + t^2y = 2ct$ has a perpendicular of $t^2x - y = 0$ passing through the origin

(ii)

$$\left. \begin{array}{l} x + t^2y = 2ct \\ t^2x - y = 0 \end{array} \right\} \Rightarrow x + t^2(t^2x) = 2ct$$

$$\therefore (1 + t^4)x = 2ct \Rightarrow x = \frac{2ct}{1 + t^4}$$

$$\therefore y = t^2 \times \frac{2ct}{1 + t^4} = \frac{2ct^3}{1 + t^4}$$

$$T\left(\underbrace{\frac{2ct}{1+t^4}}_x, \underbrace{\frac{2ct^3}{1+t^4}}_y\right)$$

$$\therefore \frac{y}{x} = t^2$$

$$y^2 = \left(\frac{2ct^3}{1+t^4} \right)^2 = \frac{4c^2(t^2)^3}{[1+(t^2)^2]^2} = \frac{4c^2\left(\frac{y}{x}\right)^3}{\left[1+\left(\frac{y}{x}\right)^2\right]^2} = \frac{\frac{4c^2y^3}{x^3}}{\left(\frac{x^2+y^2}{x^2}\right)^2} = \frac{4c^2y^3x}{(x^2+y^2)^2}$$

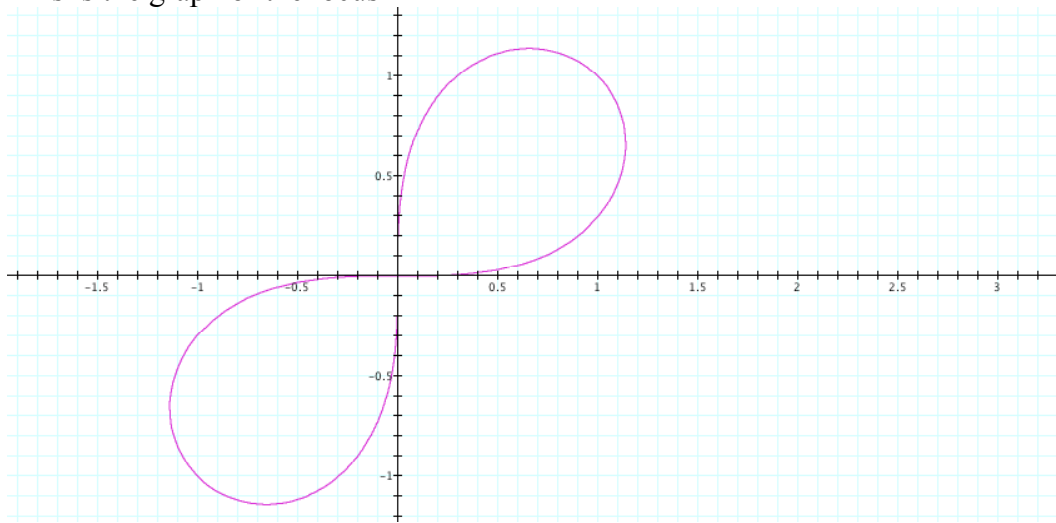
$$\therefore y^2 = \frac{4c^2y^3x}{(x^2+y^2)^2} \Rightarrow y^2 - \frac{4c^2y^3}{(x^2+y^2)^2} = 0$$

$$\therefore \frac{y^2}{(x^2+y^2)^2} [(x^2+y^2)^2 - 4c^2xy] = 0$$

$$\therefore y = 0 \text{ or } (x^2+y^2)^2 = 4c^2xy$$

$$\text{But } y = 0 \Rightarrow t = 0 \Rightarrow (0,0)$$

This is the graph of the locus



$$(d) \quad (i) \quad \text{NTP } \int_a^b f(x) dx = \int_a^b f(b-x+a) dx = \int_a^b f(b+a-x) dx$$

$$\text{LHS} = \int_a^b f(x) dx \quad \left[\begin{array}{l} u = b+a-x, du = -dx \\ x = a, u = b; x = b, u = a \end{array} \right]$$

$$= -\int_b^a f(b+a-u) du$$

$$= \int_a^b f(b+a-u) du \quad \left[\int_a^b f(x) dx = -\int_b^a f(x) dx \right]$$

$$= \int_a^b f(b+a-x) dx \quad [\text{Definite integrals are independent of pronumeral}]$$

$$= \int_a^b f(b-x+a) dx$$

$$= \text{RHS}$$

$$\begin{aligned}
\text{(ii)} \quad I &= \int_{L=a}^{L+\frac{\pi}{2}=b} \frac{A \sin(x-L) + A + B \cos(x-L) + B}{\sin(x-L) + \cos(x-L) + 2} dx \\
&= \int_0^{\frac{\pi}{2}} \frac{A \sin(u) + A + B \cos(u) + B}{\sin(u) + \cos(u) + 2} du \quad [u = x - L, du = dx] \\
&= \int_0^{\frac{\pi}{2}} \frac{A \sin\left(\frac{\pi}{2} - u\right) + A + B \cos\left(\frac{\pi}{2} - u\right) + B}{\sin\left(\frac{\pi}{2} - u\right) + \cos\left(\frac{\pi}{2} - u\right) + 2} du \quad [\text{From (i)}] \\
&= \int_0^{\frac{\pi}{2}} \frac{A \cos(u) + A + B \sin(u) + B}{\cos(u) + \sin(u) + 2} du \\
2I &= \int_0^{\frac{\pi}{2}} \frac{A \sin(u) + A + B \cos(u) + B}{\sin(u) + \cos(u) + 2} du + \int_0^{\frac{\pi}{2}} \frac{A \cos(u) + A + B \sin(u) + B}{\cos(u) + \sin(u) + 2} du \\
&= \int_0^{\frac{\pi}{2}} \left[\frac{A \sin(u) + A + B \cos(u) + B}{\sin(u) + \cos(u) + 2} + \frac{A \cos(u) + A + B \sin(u) + B}{\cos(u) + \sin(u) + 2} \right] du \\
&= \int_0^{\frac{\pi}{2}} \frac{A \sin(u) + A + B \cos(u) + B + A \cos(u) + A + B \sin(u) + B}{\sin(u) + \cos(u) + 2} du \\
&= \int_0^{\frac{\pi}{2}} \frac{(A + B)[\sin(u) + \cos(u) + 2]}{\sin(u) + \cos(u) + 2} du \\
&= \int_0^{\frac{\pi}{2}} (A + B) du \\
&= (A + B) \int_0^{\frac{\pi}{2}} 1 du \\
&= (A + B) \left[\frac{\pi}{2} - 0 \right] \\
&= (A + B) \frac{\pi}{2}
\end{aligned}$$

$$\therefore I = (A + B) \frac{\pi}{4}$$